A small mathematical notebook

Derivatives, integrals, and a useful identity

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This is a sample document for a personal notebook. The source is Markdown with LaTeX math; the PDF is typeset by Typst.

1. A derivative from first principles

Let $f(x) = x^2$. Starting with the definition,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \to 0} (2x + h) = 2x.$$

The derivative describes the local rate of change. At $x = 3$, the slope is $f'(3) = 6$.

2. An integral

The corresponding antiderivative gives

$$\int_a^b x^2 \, \mathrm{d}x = \frac{b^3 - a^3}{3}.$$

For example, $\int_0^1 x^2 \, \mathrm{d}x = 1/3$.

3. A useful identity

For vectors $u, v \in \mathbb{R}^n$, the Cauchy-Schwarz inequality states that

$$\left| \sum_{i=1}^{n} u_i v_i \right| \le \sqrt{\sum_{i=1}^{n} u_i^2} \sqrt{\sum_{i=1}^{n} v_i^2}.$$

Equality holds when the vectors are linearly dependent.

4. Keep a reproducible note

1. State the question and the assumptions.
2. Show the essential steps.
3. Include a small example that can be checked.

This PDF accompanies the sample blog article *From equations to documents*.